Method	Mean AUC	Mean T@1%FPR
No Attack	0.8418	0.4935
Simple Paraphrase	0.8588	0.2885
AdvPara (RoBERTa-Large) (N=1)	**0.5500**	**0.0740**
AdvPara (RoBERTa-Large) (N=2)	0.6632	0.1333
AdvPara (RoBERTa-Large) (N=3)	0.7266	0.1690
AdvPara (RoBERTa-Large) (N=4)	0.7593	0.1820
AdvPara (RoBERTa-Large) (N=5)	0.7788	0.2115

Table 10: Ablation on detector guidance frequency (N) during adversarial paraphrasing. Results are averaged over 8 deployed detectors. Lower AUC and T@1%FPR indicate stronger attack effectiveness.

E Failure Cases and Backfires

While our attack is highly effective, it is not always successful on every single sample, and occasional backfiring can occur—though such instances are rare. Assuming a fixed detection threshold of 0.5 for NN-based detectors and focusing on the more challenging *transfer* setting (i.e., different guidance–detector pairs), we analyzed a total of 24,000 samples.

Out of these, 5,742 samples exhibited an *increase* in detection score after paraphrasing. Among them:

- **1,773** examples were already flagged as AI-generated before paraphrasing. In these cases, the attack simply failed to help.
- **2,974** examples were classified as human both before and after paraphrasing, meaning they consistently evaded detection—thus, the attack did not worsen the outcome.
- **995** examples were originally classified as human but crossed the 0.5 threshold after paraphrasing, resulting in a *backfire* where the paraphrased text was more likely to be detected as AI-generated.

In summary:

- The attack **failed** in 1,773 cases ($\sim$7.4% of all samples).
- It **backfired** in 995 cases ($\sim$4.15% of all samples).

Despite these occasional failures, our method consistently outperforms baselines in reducing AUC and T@1%F, as demonstrated in our results. Such failures are expected in adversarial settings but do not undermine the overall robustness, effectiveness, or transferability of our approach.

F More Examples of Paraphrased Texts

Tables 12 through 18 present additional examples of original AI-generated texts along with their corresponding simple and adversarial paraphrases, the latter guided by OpenAI-RoBERTa-Large [30].

Tables 12 to 15 include examples in which both simple and adversarial paraphrases received a rating of 5. Table 16 shows instances where the simple paraphrases received a rating of 5, while the adversarial paraphrases received a rating of 4. For these cases, we also include the justification provided by GPT-4o for the score assigned. Tables 17 and 18 presents examples where neither the simple nor the adversarial paraphrases achieved a rating of 5. Justifications from GPT-4o are again included for these examples.

Across the 100 simple paraphrased texts, 76 received a rating of 5, 17 received a rating of 4, and 7 received a rating of 3. For the 100 adversarially paraphrased texts, 64 received a rating of 5, 21 received a rating of 4, 14 received a rating of 3, and 1 received a rating of 2. This shows that simple and adversarial paraphrases result in comparable text quality.

G System and User Prompt for GPT-4o Quality Rating and Win-Rate Judgment

We configure GPT-4o as a text quality rater using the system prompt shown in Figure 8, and as a win rate judge using the system prompt shown in Figure 9. The user prompt for quality rating is shown in Figure 10, and the user prompt for win rate judging is shown in Figure 11.

```
You are an expert linguist and paraphrase evaluator. Your task is to assess the quality of a paraphrased text compared to the original source text. Use the following scoring criteria:

5 - Approximately equivalent: Meaning is preserved; differences are only in wording or structure.
4 - Nearly equivalent: Meaning is mostly preserved; minor factual details differ.
3 - Somewhat equivalent: Some meaning is preserved; important details or meanings differ.
2 - Topically related: The texts are on the same topic but most meaning is lost.
1 - Not topically related: The texts are not related in topic or meaning.

Provide your final output as a JSON object in this format:
{
"score": <score from 1 to 5>,
"justification": "<brief explanation>"
}
```

Figure 8: The system prompt used to configure GPT-4o as a quality rater for evaluating the quality of paraphrased texts.

```
You are a neutral and highly capable evaluator. Your task is to compare two versions of a text and select the one that communicates the ideas more clearly, accurately, and fluently. Consider:

- Fidelity of meaning (how clearly the core message is conveyed)
- Clarity and conciseness
- Grammatical correctness and fluency
- Naturalness and appropriateness of phrasing


Give your vote solely based on quality. If one of them is a CLEAR winner, ONLY then vote for that one. Otherwise, vote for ‘tie‘.

Respond with **only one of the following**, and nothing else:
- text1
- text2
- tie
```

Figure 9: The system prompt used to configure GPT-4o as a win-rate judge for comparing the quality of simple and adversarial paraphrases.

```
Evaluate the following paraphrase using the criteria above:

Original Text:
<original_text>

Paraphrased Text:
<paraphrased_text>

What score (1 to 5) would you assign to this paraphrase, and why?
```

Figure 10: The user prompt for querying GPT-4o for the quality ratings.

Compare the following two texts and give your vote depending on meaning clarity, fluency, and overall quality. If one of them is a CLEAR winner, ONLY then vote for that one. Otherwise, vote for ‘tie‘. Respond with one of these 3 options: ‘text1‘, ‘text2‘, ‘tie‘.

Text 1:
<text1>

Text 2:
<text2>

Figure 11: The user prompt for querying GPT-4o for win rate judging.

H Detailed GPT-4o Ratings for All Detectors and Datasets

In Table 11, we report the detailed GPT-4o quality ratings (mean ± std) and the percentage of high-quality scores (ratings 4 and 5) for all paraphrased outputs across the three datasets involved. It can be observed that while adversarial paraphrasing leads to a slightly lower average quality rating compared to simple paraphrasing, the difference is not statistically significant. On average, across all datasets and guidance detectors, 87% of the adversarially paraphrased texts received a quality rating of 4 or 5.

Original Text	**Guidance Detector**	**Avg. Rating (mean ± std)**		**Rating 5&4 (in %)**	
		Simple Para.	**AdvPara**	**Simple Para.**	**AdvPara**
MAGE	mage	4.71 ± 0.61	4.54 ± 0.70	92%	88%
	radar		4.45 ± 0.80		85%
	robbase		4.54 ± 0.59		95%
	roblarge		4.48 ± 0.77		85%
KGW Watermarked MAGE	mage	4.72 ± 0.70	4.63 ± 0.63	92%	94%
	radar		4.24 ± 1.02		79%
	robbase		4.41 ± 0.83		86%
	roblarge		4.46 ± 0.81		86%
Unigram Watermarked MAGE	mage	4.71 ± 0.59	4.41 ± 0.94	95%	85%
	radar		4.21 ± 1.00		82%
	robbase		4.39 ± 0.75		86%
	roblarge		4.71 ± 4.45		93%

Table 11: GPT-4o quality ratings (mean ± std) and percentage of high-quality scores (ratings 4 and 5) for outputs rewritten using different guidance detectors. Ratings are split across simple and adversarial paraphrase scenarios.

I Computation Resources

We utilize two NVIDIA RTX A6000 GPUs to host both the paraphraser language model and the guidance AI text detector. Notably, since our approach is compatible with any instruction-tuned language model for paraphrasing and any neural network–based AI text detector, the computational requirements may vary depending on the specific models used.